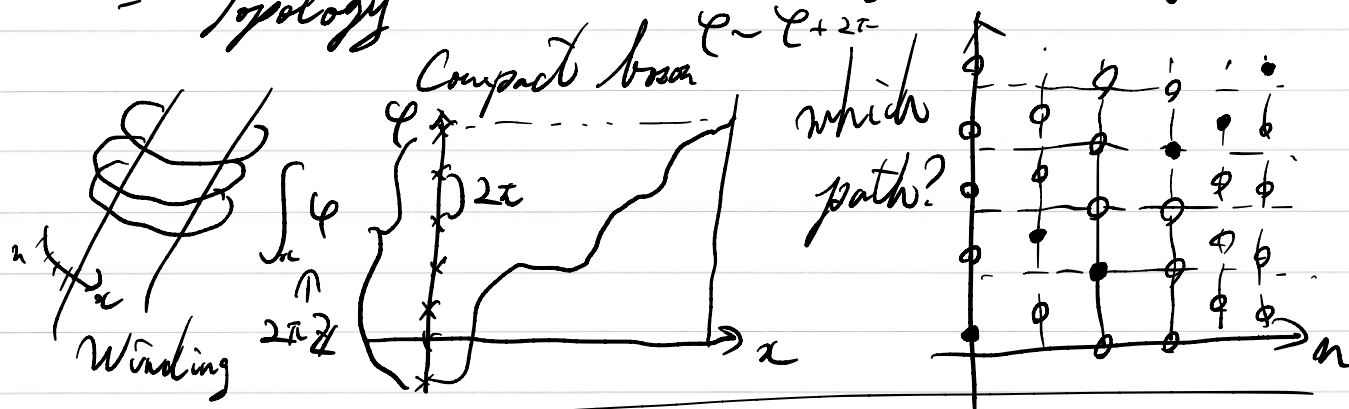


Lattice gauge theory

Department of Physics

Kyushu University We Are Renormalists

- Well-established framework of QFT
- Confinement, hadron spectrum, and scattering (HAL)
- QFT ... physics ... renormalization
- "Lattice" * gauge invariant * spacetime symmetry, continuity, etc.
- Topology



Important: continuum limit $x = na, a \rightarrow 0$
 - practically, numerical simulation at $a > 0$

~~Diagram~~ } $N \sim 128, L = 10$ fm
 nuclear scale

* MC, importance sampling
 → statistical error (# conf)

- Why correct?

What is

- * continuity, topology
- * fiber bundle of quantum phys.
- * index theorem

90s lattice fermion "overlap" Dirac op. : $\text{Ind}(D) \sim \text{ch}^{\text{QCD}} = \frac{1}{8\pi^2} \int \text{tr} F \wedge F$
 $n_+ - n_-$: K-theory (20)

